

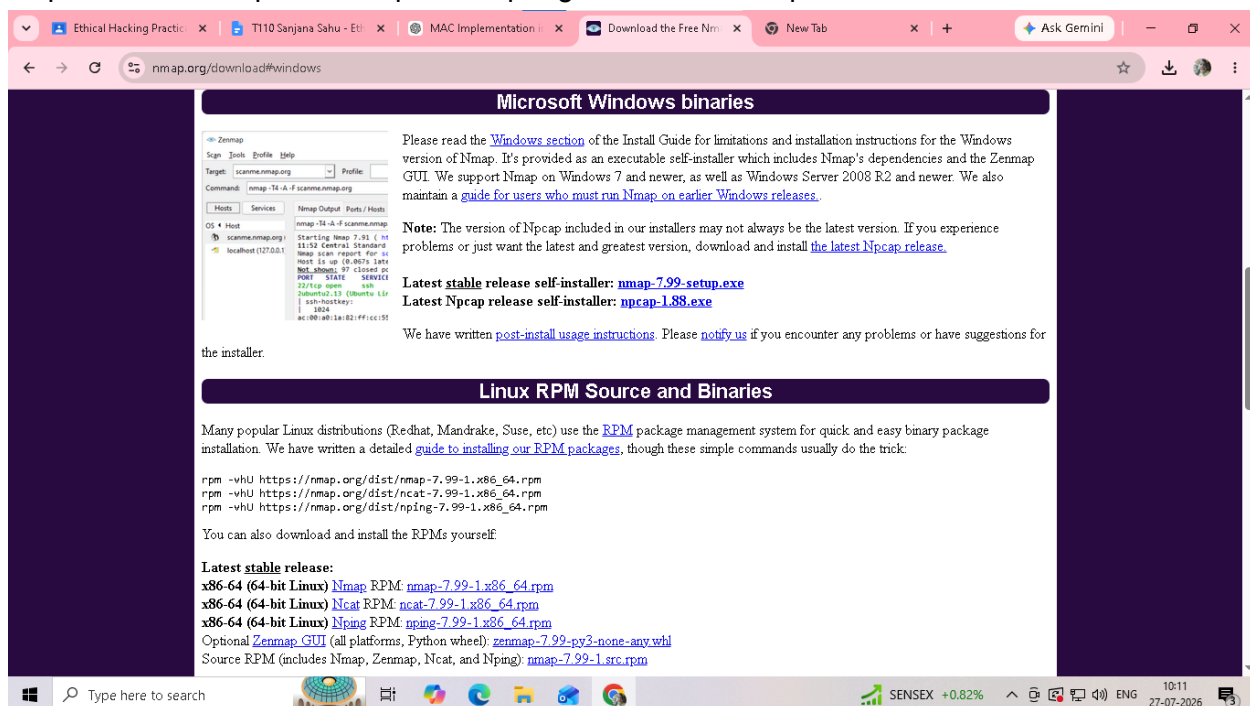
PRACTICAL NO 4

Aim:

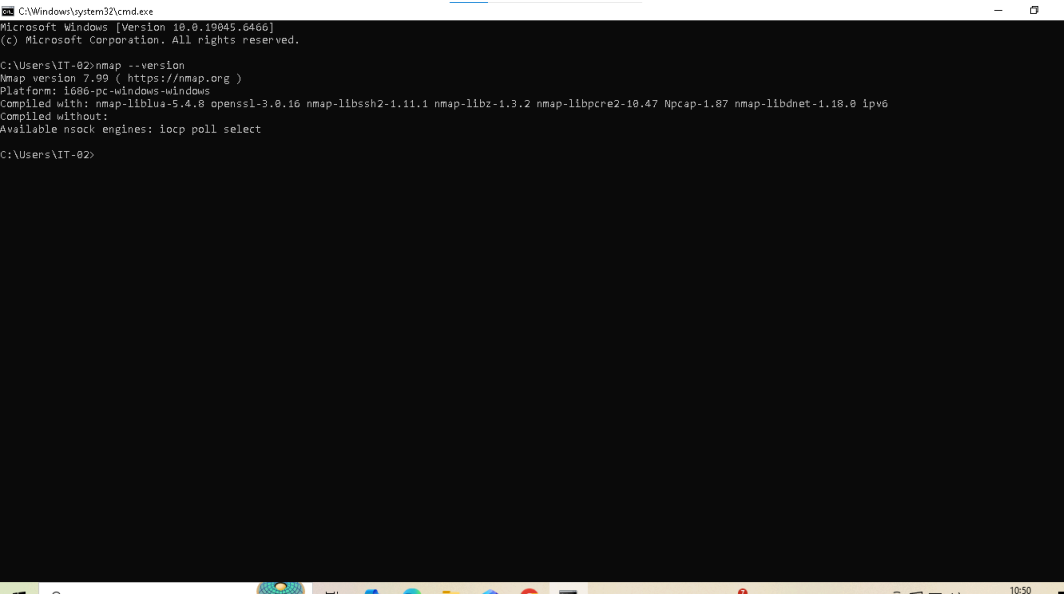
Port Scanning with NMap

- Use NMap to perform an ACK scan to determine if a port is filtered, unfiltered, or open.
- Perform SYN, FIN, NULL, and XMAS scans to identify open ports and their characteristics.
- Analyze the scan results to gather information about the target system's network services.
- Use NMap to perform an ACK scan to determine if a port is filtered, unfiltered, or open.

Step 1 : Install nmap from : <https://nmap.org/download> and open it.



1. Nmap install hua hai ya nahi check karta hai.
2. Nmap ka version dikhata hai.
COMMAND:nmap --version

A screenshot of a Windows command prompt window. The title bar at the top reads "C:\Windows\system32\cmd.exe". The window contains the following text:
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\IT-02>nmap --version
nmap version 7.99 (<https://nmap.org>)
Platform: i686-pc-windows-winsoc
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.87 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\IT-02>
The taskbar at the bottom shows the Start button, a search bar with the text "Type here to search", several application icons (Task View, Edge, File Explorer, etc.), and system tray icons including a weather widget showing "28°C Mostly cloudy", language "ENG", and the time "10:50".

1. Basic port scan karta hai.
2. Open ports aur services dikhata hai.

COMMAND:nmap scanme.nmap.org

```
C:\Users\IT-02>nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 10:53 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
179/tcp   filtered bgp
445/tcp   filtered microsoft-ds
646/tcp   filtered ldp
9920/tcp  open  nping-echo
81337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 5.67 seconds

C:\Users\IT-02>
```

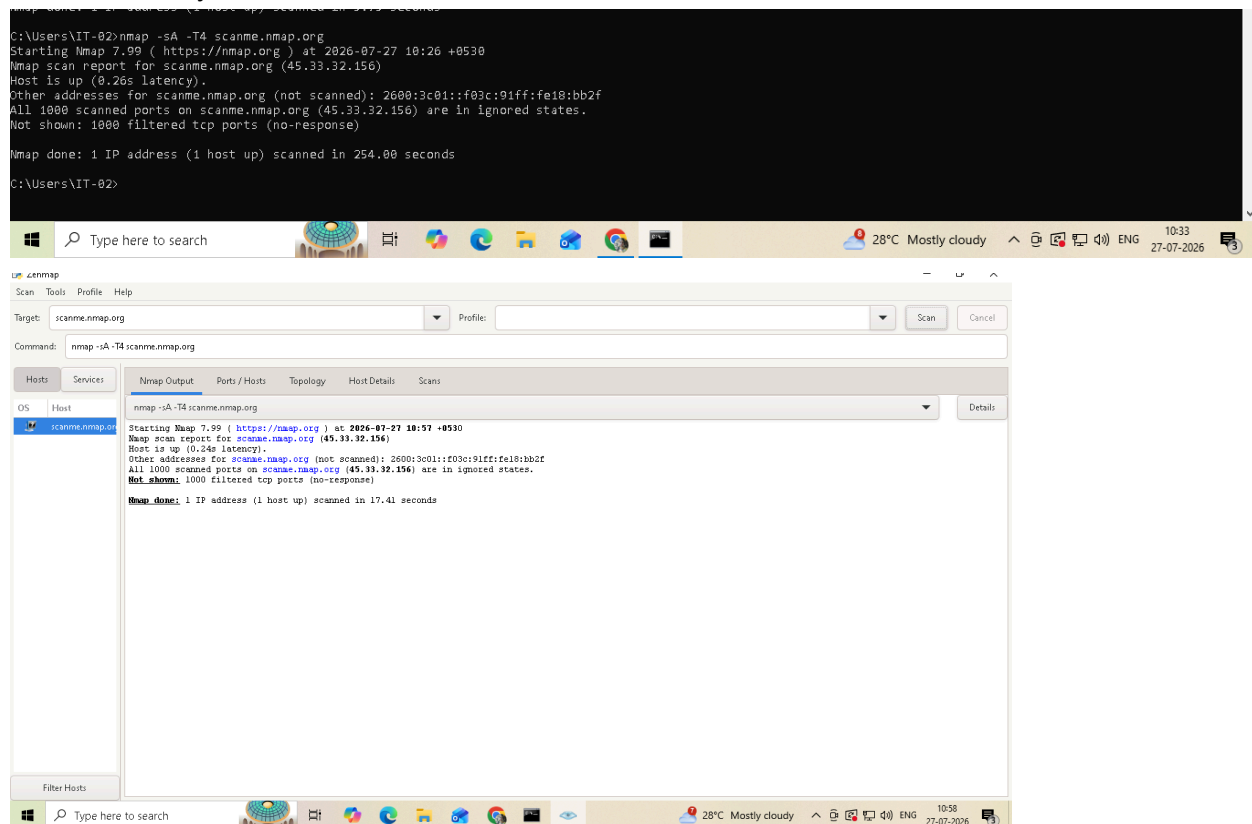
Step 2 : Type “scanme.nmap.org” to check nmap

ACK-SA (TCP ACK scan)

COMMAND: nmap -sA-T4 scanme.nmap.org

DISCRIPTION: 1. Firewall ko check karta hai.

2. Port filtered ya unfiltered hai, batata hai.



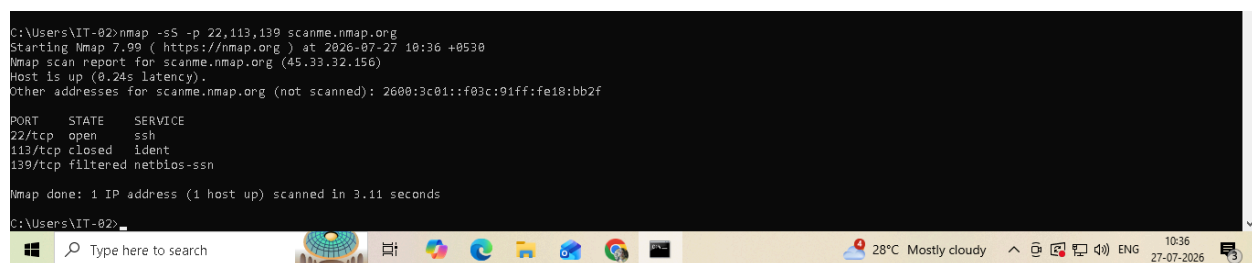
Perform SYN, FIN, NULL, and XMAS scans to identify open ports and their

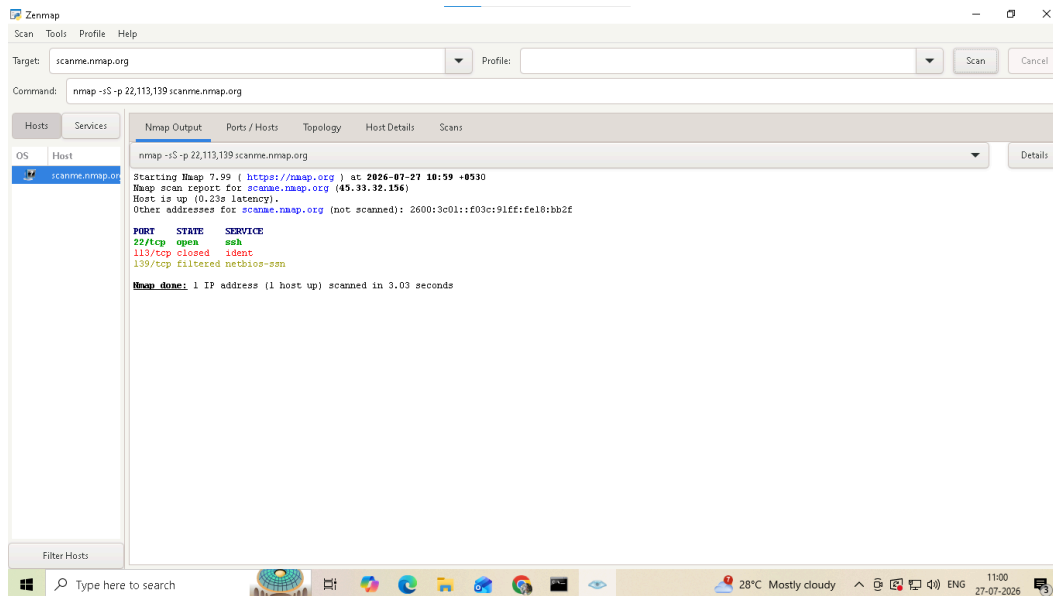
1 SYN (Stealth) Scan (-sS)

COMMAND: nmap -p22,113,139 scanme.nmap.org

DISCRIPTION::1. Open ports detect karta hai.

2. Fast aur stealth scan hota hai



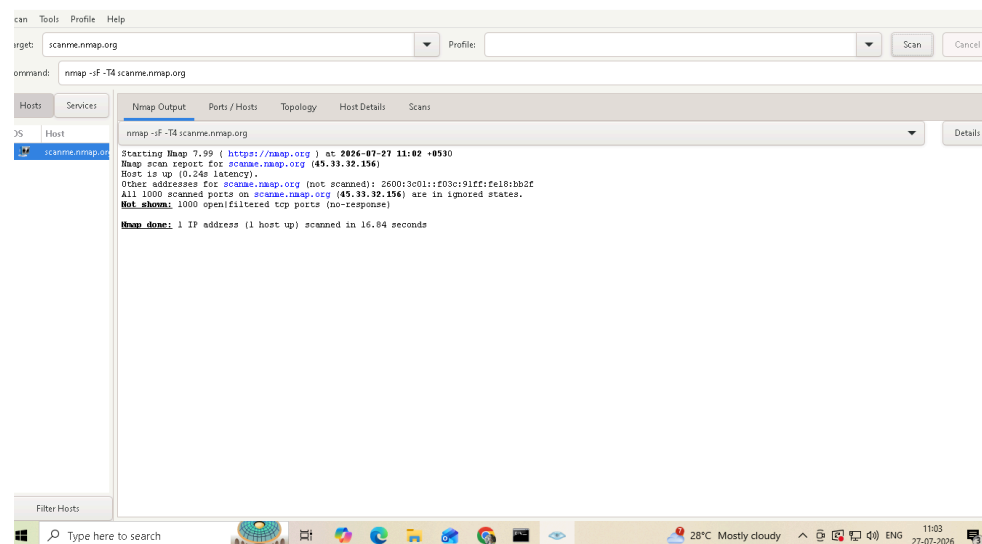
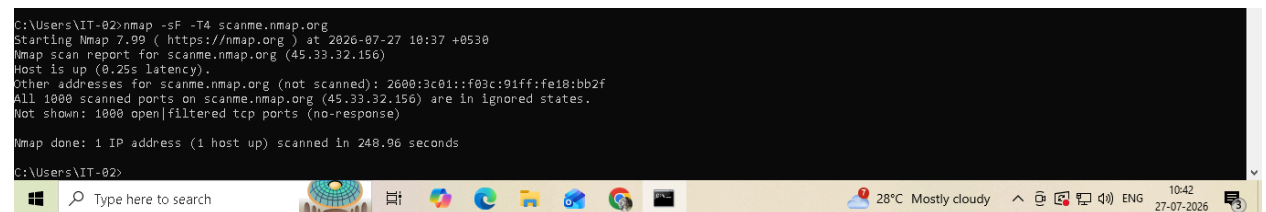


2 FIN (-sF)

COMMAND: nmap -sF -T4 scanme.name.org

DISCRIPTION: 1. FIN packet bhejkar ports check karta hai.

2. Firewall ko bypass karne me useful ho sakta hai.



3 NULL Scan (-sN)

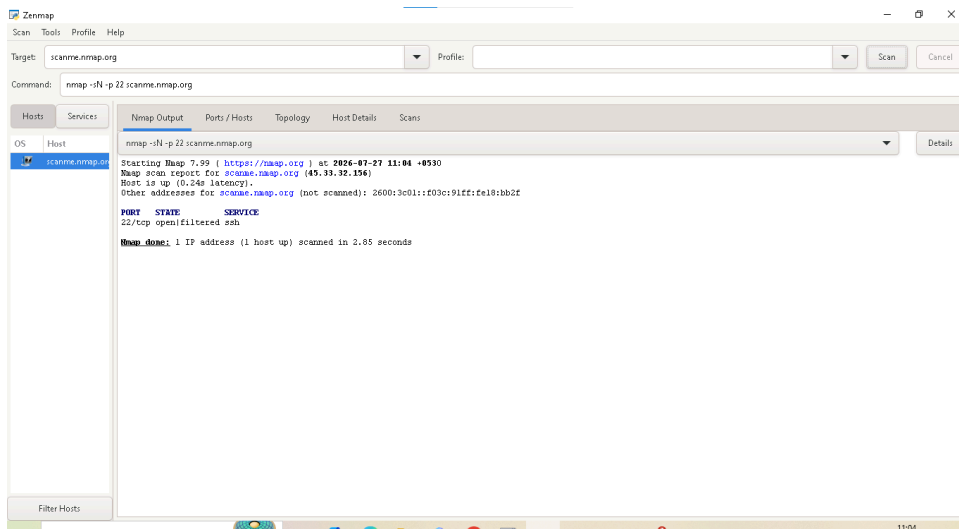
COMMAND: nmap -sN -p 22 scanme.nmap.org

- DISCRIPTION:**
1. TCP flags ke bina packet bhejta hai.
 2. Open ya filtered port identify karta hai.

```
C:\Users\IT-02>nmap -sN -p 22 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 10:43 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE      SERVICE
22/tcp    open|filtered ssh

Nmap done: 1 IP address (1 host up) scanned in 2.86 seconds
C:\Users\IT-02>
```



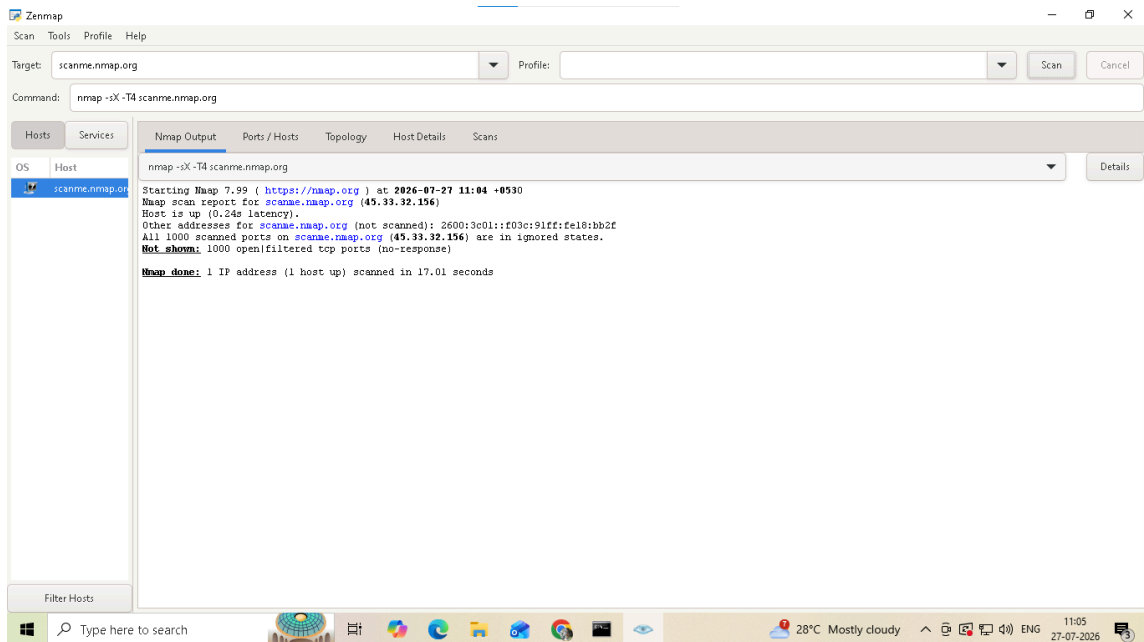
4 XMAS Scan (-sX)

COMMAND: nmap -sX-T4 scanme.nmap.org

- DISCRIPTION:**
1. FIN, PSF aur URG flags ke saath packet bhejta hai.
 2. Open ya filtered ports detect karta hai.

```
C:\Users\IT-02>nmap -sX -T4 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 10:44 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 16.93 seconds
C:\Users\IT-02>
```



Viva Questions

Q1. Nmap kya hai?

Ans: Nmap ek network scanning tool hai jo open ports aur network services ko detect karta hai.

Q2. ACK Scan ka use kya hai?

Ans: Firewall aur packet filtering check karne ke liye.

Q3. SYN Scan ko Stealth Scan kyun kehte hain?

Ans: Kyunki yeh TCP connection complete nahi karta aur jaldi scan karta hai.

Q4. FIN Scan kya karta hai?

Ans: FIN flag bhejkar open ports identify karta hai.

Q5. NULL Scan me kya hota hai?

Ans: Packet bina kisi TCP flag ke bheja jata hai.

Q6. XMAS Scan kya hai?

Ans: Packet me FIN, PSH aur URG flags set hote hain.